

1. Consider the sentence: [2]

learn this, learn that, learn all you can but also learn to unlearn.

- a. Give the count of tokens and types for the following sentence.

Tokens: 16

Types: 12

- b. Give the number of bigrams. 15

Note: No partial marking

2. For a trigram language model, find the next probable word (blank) in the sequence considering Laplace smoothing. Show the computation. [3]

Corpus: ["He is apple of my eye.", "He hit my eye."]

Vocabulary: [He, is, apple, of, my, eye, hit, ate, .]

Sequence: He ate my _____

$|Vocabulary| = |V| = 9$

For a trigram model, below are the probabilities of the next probable word with Laplace smoothing,

$P(\text{He} | \text{ate my}) = \frac{\text{count}(\text{ate my He})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{is} | \text{ate my}) = \frac{\text{count}(\text{ate my is})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{apple} | \text{ate my}) = \frac{\text{count}(\text{ate my apple})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{of} | \text{ate my}) = \frac{\text{count}(\text{ate my of})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{my} | \text{ate my}) = \frac{\text{count}(\text{ate my my})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{eye} | \text{ate my}) = \frac{\text{count}(\text{ate my eye})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{hit} | \text{ate my}) = \frac{\text{count}(\text{ate my hit})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

$P(\text{ate} | \text{ate my}) = \frac{\text{count}(\text{ate my ate})+1}{\text{count}(\text{ate my})+9} = \frac{0+1}{0+9} = 1/9$

All the probabilities are the same because the word "ate" is not present in the corpus. Hence, the next probable word can be any word present in the vocabulary because the probability of any word is the same.

One of the heuristics could be sorting the vocabulary and then picking the first word from the vocabulary. Therefore, the next predicted word is "apple".

Note:

2 marks for computation and showing it will be a tie.

1 mark for the final word based on a heuristic. Accept any heuristics that resolve the tie.

3. Give scenarios (situation, task, problem definition, etc.) when the following metric is preferred over the other and why? [2]

- a. Recall

Airport Security: Catching ALL individuals who are carrying restricted/illegal items is of paramount importance, even if we have to check each and every traveler. Thus, recall will be 100% but very low precision.

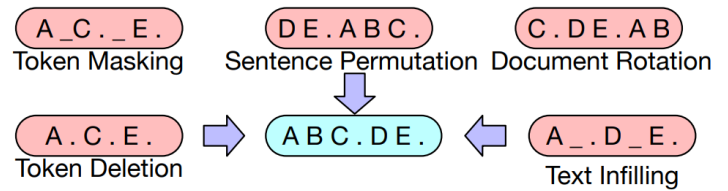
- b. Precision [2]

Spam mail identification: Precision of spam identification is important; i.e., we should mark only those mails as spam who are indeed spam. In the process, if some spam mails land in our inbox (low recall), it is not a huge problem compared to marking an important mail as spam.

Note: These are just examples. Other scenarios are also acceptable as long as they are justified. No marks without justification.

4. Differentiate between Token masking, Token deletion, and Text Infilling in BART. Use proper examples. [3]

- Token Masking: Token is masked. The model predicts the masked token.
- Token Deletion: Random tokens are deleted. The model predicts the positions of missing inputs.
- Text Infilling: Multiple tokens (span) are masked with a single masked token.



Note: 0.5 marks for definition and 0.5 marks each for examples.

5. Differentiate between local and distributed word representations. Name a technique for each. [2]

Local representation: When the info of a word is localized to only one of the dimensions of embedding. **One-hot encoding.**

Distributed representation: When the info of a word is spread/distributed to multiple dimensions of embedding. **Word2Vec, Co-occurrence vector, etc.**

6. Smoothing technique is used to avoid zero-probabilities. State the generic principle of any smoothing technique on how they avoid zero-probabilities. [2]

Steal the probability mass from words with non-zero probability and distribute it to words with zero-probability.

7. Given a phrase/sentence which can have **more than two** different interpretations depending on the context in which it is used. Mention its interpretation. [Note: Don't refer to "Achcha" from the movie PK.] [3]

Many possibilities. Will assess based on the answer. One such example could be: Public demand changes

Interpretation 1: Public demand changes, and we companies have to adapt to such changes.

Interpretation 2: Public demand changes, but does anybody listen to them?

Interpretation 3: Public demand changes have pushed many companies out of business

Note: As announced during the exam, we are not looking for a word but phrases or sentences. Zero marks if interpretations are less than three.

8. Given the following corpus with frequency, compute vocabulary till 2nd merge using word-piece tokenizer [5]

Corpus: ("hug", 10), ("pug", 5), ("pun", 12), ("bun", 4)

Initial Token Boundaries: ("h", "##u", "##g", 10), ("p", "##u", "##g", 5), ("p", "##u", "##n", 12), ("b", "##u", "##n", 4)

Initial Vocabulary Frequencies: ("b", 4), ("h", 10), ("p", 17), ("##g", 15), ("##n", 16), ("##u", 31)

Pair Frequencies: ("h", "##u") = 10, ("##u", "##g") = 15, ("p", "##u") = 17, ("##u", "##n") = 16, ("b", "##u") = 4

First Merge

Compute likelihood:

$$\text{Score}("h", "##u") = 10/10 \times 31 = 1/31$$

$$\text{Score}("##u", "##g") = 15/31 \times 15 = 1/31$$

$$\text{Score}("p", "##u") = 17/17 \times 31 = 1/31$$

$$\text{Score}("##u", "##n") = 16/31 \times 16 = 1/31$$

$$\text{Score}("b", "##u") = 4/4 \times 31 = 1/31$$

All likelihood equal. To break the tie, a heuristic should be used. For example, take the alphabetic (lexicographical) order.

Note: If students use any other convention/rule to break the tie, it's fine.

1st Merge: ("##u", "##g") → "##ug"

Vocabulary Update: ["b", "h", "p", "##g", "##n", "##u", "##ug"]

Second Merge

Updated Token Boundaries: ("h", "##ug", 10), ("p", "##ug", 5), ("p", "##u", "##n", 12), ("b", "##u", "##n", 4)

Updated Token Frequencies: ("b", 4), ("h", 10), ("p", 17), ("##n", 16), ("##u", 16), ("##ug", 15)

Updated Pair Frequencies: ("h", "##ug") = 10, ("p", "##ug") = 5, ("p", "##u") = 12, ("##u", "##n") = 16, ("b", "##u") = 4

Compute likelihood:

$$\text{Score}("h", "##ug") = 10/10 \times 15 = 1/15$$

← Max

$$\text{Score}("p", "##ug") = 5/17 \times 15 = 1/51$$

$$\text{Score}("p", "##u") = 12/17 \times 16 = 3/68$$

$$\text{Score}("##u", "##n") = 16/16 \times 16 = 1/16$$

$$\text{Score}("b", "##u") = 4/4 \times 16 = 1/16$$

2nd Merge: ("h", "##ug") → "hug"

Vocabulary Update: ["b", "h", "p", "##g", "##n", "##u", "##ug", "hug"]

Note: Based on the heuristic, vocabulary may look different.

9. Can we use MLM for word2vec CBOW? Justify.

[2]

Yes, it is possible. The objective of both is to predict a word based on given (previous / future) context. In CBOW, we can mask a token that can be predicted at the output.

Note: Use your judgement for the justification.

10. What do we mean by the phrase "GPT is auto-regressive in nature"? State GPT objective functions in equations. [4]

Auto-regressive means that tokens are predicted one token at a time and we do not use future context to predict any token.

GPT has two objectives:

a. Language Model:
$$L_1(\mathcal{U}) = \sum \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta)$$

b. Supervised classification task:
$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y | x^1, \dots, x^m).$$

End